

Beyond Functionality: Olfactory Cues for Affective Accessibility in Collaborative Social XR

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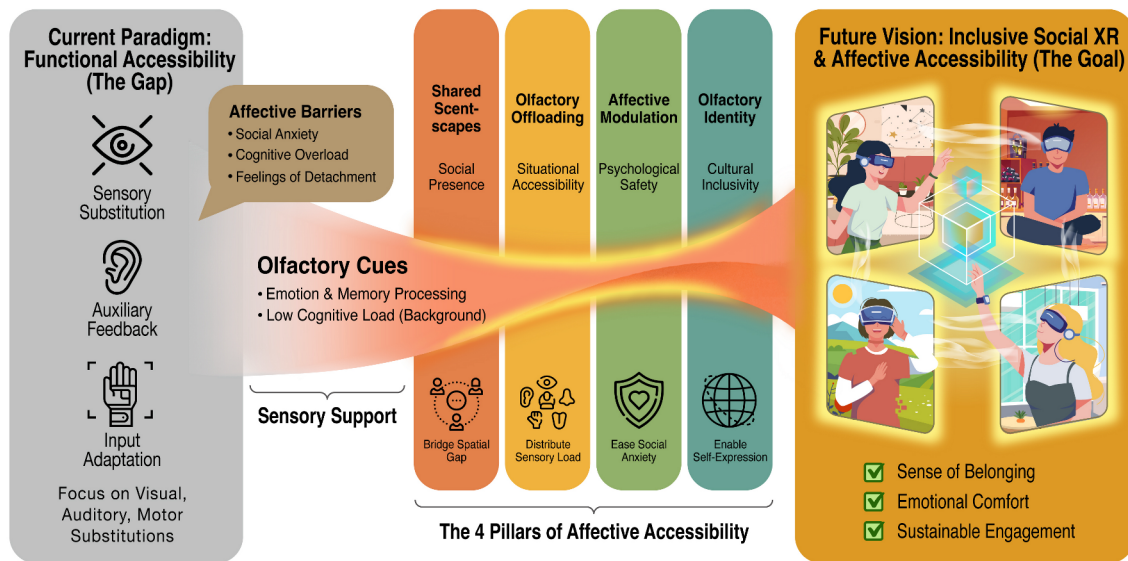


Figure 1: Conceptual framework illustrating the transition from functional accessibility to affective accessibility in Social XR through olfactory cues.

ABSTRACT

As eXtended Reality (XR) platforms increasingly support global collaboration, prevailing accessibility frameworks remain largely anchored in a functional paradigm that emphasizes visual, auditory, and motor substitutions. This focus, however, often overlooks what we term affective accessibility, used here to describe the psychological and emotional ease with which diverse users can enter, remain, and meaningfully engage in shared immersive social spaces. This position paper argues that olfactory cues, given their close association with affective experience, represent a promising yet under-explored channel for supporting emotional inclusion in Social XR. Rather than serving solely as immersive enhancements, integrated scent-scapes may function as subtle modulators of social anxiety, support shared affective conditions, and enable cultural expression within collaborative virtual environments. By examining olfactory cues through the lenses of situational accessibility, psychological safety, and inclusive design, this paper articulates a conceptual shift toward a holistic, multi-sensory approach that foregrounds users' emotional comfort, sense of belonging, and capacity for meaning-

ful participation.

Index Terms: Social XR, Olfactory Cues, Affective Accessibility, Inclusive Design, Psychological Safety

1 INTRODUCTION

Extended reality (XR) technologies, including Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR) are transforming how people engage in remote collaboration within digital spaces [18, 35, 41, 9, 1]. As XR finds broader application in remote work, educational training, and cross-regional collaboration, researchers have increasingly recognized that functional usability alone does not guarantee seamless user participation [24, 22]. In this context, accessibility is often understood through a functional paradigm, which focuses on supporting users with sensory or motor impairments through alternative feedback channels and adaptive input mechanisms [27, 45]. However, we argue that truly inclusive design requires a shift toward what we term affective accessibility, defined here as the psychological and emotional ease with which diverse users can enter, remain, and meaningfully engage in shared immersive social spaces.

To date, research into XR accessibility has primarily focused on supporting users with sensory or motor impairments through alternative feedback channels and adaptive input mechanisms [27, 45]. However, this approach largely interprets participation barriers as stemming from physical or sensory deficits, leaving psychological states and emotional experiences less systematically incorporated into design considerations. In practical Social XR collaboration,

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users often encounter implicit barriers that extend beyond functional limitations, including social anxiety [22], information overload [44], and feelings of detachment in cross-cultural communication [21]. Even when a system achieves high functional usability, these affective factors can determine whether users successfully enter collaborative spaces, sustain their presence, or remain meaningfully engaged [31, 30].

Consider, for instance, a user joining a virtual team meeting who experiences heightened self-consciousness when their avatar movements become visible to all participants [6, 33], or an international collaborator who feels culturally isolated when the virtual environment reflects design conventions unfamiliar to their background [17]. Similarly, users with social anxiety may find the persistent visibility of their engagement status, whether they are speaking, gesturing, or remaining silent, to be a source of ongoing stress rather than a facilitator of connection [22]. These barriers are not addressed by conventional accessibility solutions focused on input and output modalities; they require attention to the affective dimensions of collaborative experience.

Building on these observations, this paper introduces affective accessibility as a conceptual lens for examining whether users can not only operate in Social XR systems, but also comfortably participate in collaborative experiences. We position olfactory cues as a key design resource for supporting this goal, as olfaction is closely linked to emotional and memory processing, enabling it to influence users' affective experiences with minimal cognitive load [11, 13]. The integration of olfaction should therefore be understood not merely as an immersive enhancement, but as a strategy for fostering emotional inclusion.

Our central position is that functional usability is necessary but insufficient for inclusive Social XR; affective accessibility, achieved in part through olfactory design, is essential for enabling diverse users to participate comfortably and sustainably. To substantiate this argument, this paper develops its analysis across four interrelated dimensions, as illustrated in Figure 1:

1. **Shared scent-scapes and social presence:** Using synchronized ambient scents to bridge physical distance and enhance the sense of co-presence.
2. **Offloading and situational accessibility:** Distributing environmental cues to the olfactory channel to alleviate cognitive saturation in visual and auditory modalities.
3. **Affective modulation and psychological safety:** Employing ambient scents to foster an atmosphere of trust and reduce social evaluation pressure.
4. **Olfactory identity and cultural inclusivity:** Providing a low-pressure, non-verbal pathway for personal and cultural expression.

2 FROM FUNCTIONAL TO AFFECTIVE ACCESSIBILITY

Current XR accessibility research follows a paradigm of functional accessibility: compensating for users' visual, auditory, or motor limitations through sensory substitution, auxiliary feedback, and input adaptation [43, 27, 45, 42, 2]. This approach encompasses a range of established techniques, including spatial audio cues for users with visual impairments [8], closed captioning and visual prompts for deaf or hard-of-hearing users [42], and adaptive controllers (e.g., Xbox Adaptive Controller) for those with motor impairments. While this paradigm constitutes the foundation for accessible XR systems, its limitations become increasingly apparent in Social XR scenarios where collaboration and social interaction are central [27, 24].

The concept of "operating a system", understood as successfully navigating interfaces, executing commands, and completing

discrete tasks, does not guarantee sustained participation in collaborative processes. Functional usability alone cannot address the social exposure pressures and emotional discomfort inherent in collaborative contexts [22, 38]. Research has documented that users in multi-user virtual environments experience heightened self-consciousness, evaluation anxiety, and social fatigue even when systems are technically accessible [6]. These affective barriers, namely the psychological and emotional obstacles that impede comfortable participation, represent a distinct category of accessibility challenges that functional approaches leave unaddressed.

We therefore introduce **affective accessibility** as a conceptual extension of traditional accessibility frameworks. We define affective accessibility as: *the degree to which collaborative environments enable users to psychologically and emotionally enter, remain within, and meaningfully engage in shared immersive social spaces*. This concept builds on established work in psychological safety [5], situational impairment [25], and social presence research [3, 30]. Unlike task-oriented design goals, affective accessibility regards psychological safety, social presence, and respect for diverse communication styles as prerequisites for collaborative experience rather than as optional enhancements. A system that is functionally complete yet induces anxiety or alienation may therefore remain effectively inaccessible for sustained collaboration [22].

It is important to distinguish affective accessibility from related concepts in Human-Computer Interaction (HCI). While user experience (UX) encompasses overall satisfaction and emotional design focuses on eliciting positive responses to products [29], affective accessibility specifically concerns the *preconditions* for participation, whether users can psychologically enter and remain within collaborative spaces in the first place. Examples of affective barriers include: social evaluation anxiety triggered by visible avatar behaviors [6, 33], cultural alienation from unfamiliar environmental design conventions [21], cognitive overload from simultaneous task and social demands [44], and persistent self-monitoring stress from engagement visibility [22].

Olfaction emerges as a promising channel for supporting affective accessibility due to several distinctive characteristics. First, it operates primarily at the experiential background level without requiring active decoding [14, 39]. Second, it is closely intertwined with emotional and memory processing through direct neural pathways to the limbic system, bypassing the thalamus unlike other sensory modalities [7, 36, 11]. Third, research demonstrates that ambient scents can influence affective states with minimal cognitive load, reducing stress, enhancing mood, and improving task performance in demanding environments [37, 10]. These characteristics position olfaction as a viable candidate for supporting emotional stability and participatory comfort within collaborative environments.

3 FOUR PILLARS OF OLFACTORY INCLUSION

Building on the dimensions introduced above, we articulate four pillars by mapping documented affective barriers in Social XR to key properties of olfactory perception. Specifically, shared scent-scapes address sensory disconnect [3], olfactory offloading responds to cognitive overload [44], olfactory modulation addresses evaluation pressure and psychological safety [5], and olfactory identity supports cultural expression [21].

3.1 Shared Scent-scapes for Social Presence

Although Social XR can simulate co-presence through virtual avatars and spatial audio [34, 20], participants often experience a sense of separation stemming from inconsistent sensory backgrounds: while users share visual and auditory cues in virtual space, they remain immersed in their distinct physical environments. This "digital vacuum" undermines the subjective experience of being co-located with others.

Shared scent-scapes offer an alternative approach. In collaborative settings, olfaction can function as a continuously present environmental cue rather than an interactive channel focused on information transmission. By introducing consistent olfactory cues during collaboration, scent may serve as a sensory reference that bridges physical spatial divides, helping participants establish a more unified sensory background. This background-level consistency can mitigate spatial discontinuity without imposing additional attentional demands.

From a design perspective, shared scent-scapes might take several forms. Neutral ambient scents, such as subtle woody or green notes, could establish a consistent “virtual room” atmosphere without imposing specific cultural associations. Alternatively, context-specific scents might be introduced to mark different phases of collaboration, such as a fresh citrus note signaling the beginning of a brainstorming session. The key design principle is that such scents should remain at the periphery of attention, supporting rather than demanding conscious engagement.

From an inclusive design perspective, the value of shared scent-scapes lies in extending the construction of co-presence beyond visible behaviors to more fundamental conditions of sensory co-existence, achieved through low-intrusive and non-verbal means.

3.2 Olfactory Offloading for Situational Accessibility

Accessibility challenges in Social XR are not always rooted in users’ long-term ability differences. They often emerge situationally under conditions of high cognitive demand [43, 25]. When users must simultaneously track shared content, interpret multi-party audio, and monitor social cues, visual and auditory channels can quickly reach cognitive saturation [40].

Traditional accessibility solutions address perceptual limitations by adding explicit cues such as visual alerts or audio notifications [27]. However, when sensory channels are already saturated, introducing additional information tends to compound rather than alleviate overload. Addressing situational accessibility therefore requires distributing cues across sensory channels instead of concentrating them within the same modalities.

Olfaction offers such a possibility because scent operates in the perceptual background and does not require focused attention [14]. This enables what we term *olfactory offloading*: using olfaction not to add new content, but to transfer environmental cues away from saturated visual and auditory channels. For example, transitions between collaborative phases can be indicated through subtle ambient scent shifts rather than visual pop-ups. In this way, users can perceive changes without processing additional explicit notifications. By accommodating individual differences in attention capacity and stress tolerance, this approach can reduce invisible participation barriers under high cognitive load.

3.3 Olfactory Modulation for Psychological Safety

Psychological safety, defined as the shared belief that a team is safe for interpersonal risk-taking [5], is critical for collaborative engagement. In Social XR, this sense of safety can be undermined by the heightened visibility of user behaviors, as avatar movements, spatial positions, and engagement status are continuously observable by all participants [6, 22]. While such visibility enhances social presence, it may also amplify self-monitoring and evaluation anxiety, particularly for users who are already socially hesitant.

Psychological safety cannot be engineered through a single feature; rather, it emerges from the overall affective atmosphere of the collaborative environment [5]. In fast-paced multi-user XR settings, relying on explicit interventions such as verbal check-ins or behavioral prompts to manage social dynamics may add to users’ cognitive load. An alternative approach is to shape the environmental backdrop in ways that subtly support emotional regulation without demanding additional attention.

Olfactory cues offer this possibility. Prior research has shown that certain scents can reduce anxiety and promote relaxation, such as lavender [19], while others support alertness and cognitive performance, including peppermint and rosemary [26, 15]. In collaborative XR contexts, scent selection could therefore be tailored to interactional demands, with calming scents applied during high-stakes negotiations or conflict-sensitive discussions and more energizing scents used during creative brainstorming. Because olfaction primarily operates at the environmental background level [14], this form of affective modulation can shape users’ emotional states without introducing explicit notifications or additional social demands.

This approach is particularly valuable for users with diverse social styles, including those who require more time to engage, prefer to observe before participating, or experience discomfort in high-visibility settings. By embedding emotional support within the environment itself, olfactory design can lower participation thresholds without requiring users to explicitly signal their needs.

3.4 Olfactory Identity and Cultural Representation

In global Social XR collaboration, inclusivity extends beyond functional participation to whether users feel their identities are recognized. While avatar customization has advanced significantly [16, 6], visual identity expression remains constrained by predefined assets and implicit design biases, as users can only choose from options provided by designers.

Olfaction offers a different pathway for self-expression. Scents carry cultural and personal meanings, as specific spices, flowers, or environmental notes often evoke home, heritage, or memory [11, 23], without requiring explicit verbal explanation. This makes olfaction suitable for implicit identity expression, allowing users to communicate aspects of themselves through ambient presence rather than declarative statements.

A concrete implementation could involve user-defined scent profiles. Using personal olfactory display devices, namely compact scent-delivery hardware kept in users’ physical environments [28, 32], individuals could select ambient scents that accompany their virtual presence. When entering a shared virtual space, each user’s device would emit the chosen scent locally, while the system could support basic coordination strategies, such as limiting overlap or managing transitions when users interact closely. These profiles may draw from culturally meaningful associations without requiring users to explicitly label or explain their background.

This approach allows diverse identities to coexist through sensory juxtaposition rather than visual categorization. Users can express cultural connection without the pressure of being labeled or explaining themselves, thereby supporting more sustainable forms of cross-cultural collaboration.

4 CHALLENGES AND ETHICAL CONSIDERATIONS

Although olfactory cues offer a new design perspective for affective accessibility, their integration must balance technological potential with ethical risks to ensure that mitigating existing barriers does not introduce new forms of exclusion.

Sensory equity presents a primary challenge. Given significant individual variations in olfactory sensitivity, perceptual ability, and physiological responses [4, 12], olfactory cues should not be positioned as essential or exclusive interaction channels. They are better utilized as augmentative offloading, ensuring that core functionalities and participation requirements remain intact even in the absence of olfactory perception. Users with anosmia or hyposmia must not be disadvantaged by olfactory-dependent designs.

User agency and psychological boundaries constitute another critical consideration. Because olfaction often influences experiential atmosphere at a subconscious level, its deployment must avoid being perceived as emotional manipulation. Within collaborative

XR contexts, scent should be regarded as a supportive environmental element rather than a behavioral steering or emotional management tool. Clarifying this positioning helps preserve user autonomy and trust.

Technical standardization poses practical obstacles. Unlike audiovisual modalities, olfactory displays lack established standards for scent representation, delivery, and cross-platform compatibility. The development of shared protocols for olfactory integration in XR platforms will be necessary for widespread adoption.

Scent conflicts in shared spaces require careful consideration. When multiple users with different scent preferences or cultural associations share a virtual environment, conflicts may arise. Design solutions might include negotiated shared scents, personal olfactory “bubbles,” or systems that blend individual preferences into harmonious combinations.

Overall, the potential for olfactory application in Social XR must be evaluated alongside technical feasibility, individual variability, and ethical boundaries. Only through transparency and restraint can olfaction serve as a sustainable, multi-sensory design resource that contributes value to inclusive collaborative environments.

5 CONCLUSION AND FUTURE WORK

As Social XR evolves into an important platform for global collaboration, the core challenge of inclusive design is shifting from technical connectivity to emotional engagement. This paper has argued that functional usability alone does not guarantee equal participation in immersive collaboration. Truly inclusive collaborative experiences depend on affective accessibility, which ensures that users can smoothly enter, remain, and sustain engagement at emotional and psychological levels.

We have repositioned olfaction as a design resource supporting affective accessibility rather than merely a sensory appendage for immersion enhancement. Through discussions of social presence, situational accessibility, psychological safety, and cultural identity representation, this paper has demonstrated olfaction’s potential as an environmental cue for lowering participation barriers, alleviating collaborative pressures, and supporting diverse engagement modalities.

Our aim is not to replace established audiovisual interaction paradigms but to introduce olfaction as a complementary dimension within a holistic multi-sensory environment. We call upon the Social XR research and design community to prioritize emotional experiences and sensory conditions within collaborative settings alongside concerns for interaction efficacy and functional accessibility.

Future research should address several open questions. First, empirical studies are needed to validate the effects of specific olfactory parameters, including scent type, intensity, and duration, on affective states in collaborative XR contexts. Second, design frameworks must be developed to guide culturally sensitive scent integration that respects diverse olfactory associations while avoiding stereotyping. Third, investigations into individual differences in olfactory perception and their implications for inclusive design will be essential. Finally, longitudinal studies examining how olfactory cues influence sustained participation and community formation in Social XR would provide valuable insights for practitioners. By pursuing these directions, the field can move toward collaborative XR environments where users from all backgrounds feel seen, respected, and empowered to engage sustainably.

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